THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE CHAPTER TEST SET B

Chapter 10: **Light: Mirrors and Lenses** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A ray of light makes an angle of 55° with the normal to a plane mirror. The angle of reflection is: 1

(a) 35° **(b)** 110° **(c)** 55° **(d)** 125°

2. The bouncing back of light from a surface is called: 1

(a) dispersion **(b)** refraction **(c)** absorption **(d)** reflection

3. If a ray falls along the normal, the angle of reflection is: 1

(a) 0° **(b)** 180° **(c)** 45° **(d)** 90°

4. A spherical mirror whose reflecting surface curves inward is: 1

(a) plane mirror **(b)** lens **(c)** concave mirror **(d)** convex mirror

5. The image formed by a convex mirror is always: 1

(a) erect and diminished **(b)** inverted and enlarged
(c) inverted and diminished **(d)** erect and enlarged

6. When an object is close to a concave mirror the image is: 1

(a) no image **(b)** inverted and diminished
(c) erect and enlarged **(d)** erect and diminished

7. A convex lens is thicker at the: 1

(a) centre **(b)** bottom only **(c)** edges **(d)** top only

8. A magnifying glass is a: 1

(a) convex mirror **(b)** plane mirror **(c)** convex lens **(d)** concave lens

9. Which can form a real image on a screen? 1

(a) convex mirror **(b)** concave lens **(c)** convex lens **(d)** plane mirror

10. The surface of a spoon that bulges outward acts like a: 1

(a) plane mirror **(b)** lens **(c)** convex mirror **(d)** concave mirror

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. Differentiate concave and convex mirrors. 2

12. State one use of a concave mirror and one of a convex mirror. 2

13. Why are convex mirrors used as rear-view mirrors? 2

14. What is lateral inversion? 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 30–40 words each.

15. Describe how a solar cooker makes use of a mirror. 3

16. Describe the image formed by a convex lens when the object is close and far. 3

17. Distinguish between a convex lens and a concave lens by looking at text through them. 3

OR

Explain with an activity why a pencil appears bent in a glass of water. 3

18. State three uses of lenses. 3

OR

State three differences between reflection and refraction. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 80–100 words each.

19. Explain convex and concave lenses, the type of image each forms and one use each. 5

OR

Describe the ray diagram (in words) for reflection from a plane mirror and mark the normal, incident ray, reflected ray, angles i and r. 5

20. Three mirrors (plane, concave, convex) are placed before a graph sheet. How will you identify each from the image? 5

OR

A woman walks towards a large concave mirror. Describe what she sees and explain why. 5

END OF QUESTION PAPER Total: 40 Marks

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